

ScriptureRAG: Complete 8-Phase Interview Guide

A Clear, Step-by-Step Breakdown of All 8 Phases + Web UI (Simple English)

1. Project Summary in Simple Words

ScriptureRAG is an authentic AI counselor for the **18 Hindu Mahapuranas** (Vishnu, Bhagavata, Shiva, Garuda, Padma, Agni, etc.). Unlike normal ChatGPT which invents fake verses, ScriptureRAG uses **RAG (Retrieval-Augmented Generation)** across 8 specialized phases to find the real book page and cite exact references (e.g., *[Bhagavata Purana, Canto 1, Page 85]*).

2. Master Table of All 8 Project Phases (+ Frontend UI)

Phase	Phase Name & File	Algorithms & Models Used	What it Does & Why it is Best
Phase 1	Environment & Data Setup (requirements.txt, .env)	Python 3.10+, PyPDF2, Kaggle Dataset	Sets up 123 Motilal Banarsidass (MLBD) Mahapurana PDFs and 1,404 Kaggle structured stories.
Phase 2	Master Preprocessing (src/preprocessing.py)	Sliding Window Chunking (800 chars / 150 overlap)	Cuts books into 10,582 clean scripture chunks with unique IDs (MLBD_{FILE}_C{INDEX}) to stop data loss.
Phase 3	Two-Layer Input Guardrail (src/guardrails.py)	Fast Regex (<1ms) + Gemini Flash Classifier	Blocks hacking, coding questions, and stocks in 1ms. Layer 2 allows real spiritual questions like stress or grief.
Phase 4	Pre-Retrieval Expansion (src/query_processor.py)	Word-Boundary Regex Matcher (<code>\b{word}\b</code>)	Translates everyday user words into Sanskrit scripture names without false matches (e.g. 'sin' in 'business').
Phase 5	Semantic Vector Search (src/vector_db.py)	BAAI/bge-base-en-v1.5 (768-d) + ChromaDB	Searches 10,582 book paragraphs in 0.01 seconds using local Cosine Similarity. 100% free forever.
Phase 6	Cross-Encoder Re-Ranking (src/reranker.py)	cross-encoder/ms-marco-MiniLM-L-6-v2	Evaluates full token-level cross-attention to re-rank Top-6 candidates down to the Top-3 most accurate passages.
Phase 7	Grounded LLM Generation (src/llm_engine.py)	Google Gemini 3.6 Flash (T = 0.2, 2048 tokens)	Synthesizes structured wisdom (Counsel, Evidence, Dharmic Principle) with strict [Purana, Page] citations.
Phase 8	Output RAGAS Guardrail (src/validation.py)	RAGAS Mathematical Entailment Formula	Audits Faithfulness ($F \geq 0.85$), Answer Relevance & Citations. Proves mathematically that the answer is true.
UI / App	ChatGPT-Style Frontend (frontend/app.py)	Streamlit + HTML5 Web Speech API (Mic)	Minimalist dark-mode chatbot where users can type or click the microphone to speak questions directly.

3. Detailed Explanation of All 8 Phases: Why & How Each Works

Phase 1 & 2: Canonical Datasets & Master Preprocessing

- **What we did:** Gathered 123 Motilal Banarsidass (MLBD) translation volumes of the 18 Mahapuranas + 1,404 Kaggle story rows.
- **Chunking Logic:** 800 characters per chunk with 150 characters overlap. This keeps complete Sanskrit verses and English commentary intact.
- **Total Saved:** 10,582 chunks saved in data/preprocessed_master_chunks.json.

Phase 3: Two-Layer Hybrid Input Guardrail (src/guardrails.py)

- **Layer 1 (Regex <1ms):** Strips hidden unicode, decodes Base64, and stops coding/finance prompts without spending API credits.
- **Layer 2 (Semantic Gemini):** Understands metaphors (e.g. 'my life has no database of joy' is allowed for Puranic guidance, but 'write SQL database code' is blocked).

Phase 4: Pre-Retrieval & Canonical Query Expansion (src/query_processor.py)

- **What it does:** Maps everyday English questions to canonical scripture concepts (e.g. 'Krishna' → 'Lord Vishnu', 'anxiety' → 'Moha / Shoka').
- **Word-Boundary Regex:** Uses `\b{word}\b` so it never matches letters inside other words (e.g. avoids matching 'sin' in 'business').

Phase 5: Dense Vector Retrieval & ChromaDB (src/vector_db.py)

- **Embedding Model:** BAAI/bge-base-en-v1.5 (768 dimensions). One of the top open-source embedding models on Hugging Face.
- **Database:** Local **ChromaDB** (Cosine distance). Fast search across 10,582 chunks in 0.01 seconds without cloud hosting costs.

Phase 6: Cross-Encoder Precision Re-Ranking (src/reranker.py)

- **Model:** cross-encoder/ms-marco-MiniLM-L-6-v2.
- **Why it is best:** Vector search finds 6 general matches. The Cross-Encoder reads both the question and paragraph together to score exact word-by-word relevance, picking the **Top 3 most accurate passages**. This removes noise and prevents the AI from getting confused.

Phase 7: Grounded LLM Generation Engine (src/llm_engine.py)

- **Model:** Google Gemini 3.6 Flash with **Temperature = 0.2** and **2,048 max output tokens**.
- **Structure:** Generates in 3 sections: (1) Direct Counsel, (2) Holy Scripture Evidence with citations [Purana, Page X], (3) Moral Duty / Dharmic principle.

Phase 8: Output Guardrail & RAGAS Verification (src/validation.py)

- **What it does:** Evaluates the generated answer against the retrieved passages to ensure **Faithfulness ≥ 0.85 (85%)**.
- **Verification Badge:** Appends **Verified Authentic Scripture Counsel | RAGAS Faithfulness: 1.00 (PASS)** to certify authenticity.

4. Frontend Web Interface (UI Tools & Technologies)

The web user interface was built to provide an ultra-clean ****ChatGPT dark-mode experience**** with voice interaction:

UI Component	Technology Used	How it Works in Simple Words
Web Framework	Streamlit (Python)	Pure Python web framework that runs all 8 RAG phases locally with high speed and zero complex JS build steps.
ChatGPT Dark Theme	Custom CSS Styling	Matches ChatGPT's official dark interface (#212121 background, #2f2f2f rounded chat bubbles, clean typography).
Voice Input (Mic)	HTML5 Web Speech API (webkitSpeechRecognition)	When the user clicks 'Speak Question' , the browser listens in Chrome/Edge, converts voice to text, and sends it with zero API cost .
Conversation State	Streamlit Session State (st.session_state)	Stores conversation history in memory and provides a 'New Chat' button to start a new inquiry anytime.

5. Top Viva & Interview Questions (Ready-to-Speak Answers)

Q1: Explain the flow of your project in 1 minute.

Answer: 'ScriptureRAG has 8 phases: Phase 1 & 2 preprocessed 123 Puranic volumes into 10,582 chunks. Phase 3 filters invalid queries via a 2-layer guardrail. Phase 4 expands queries with Sanskrit entities. Phase 5 uses BGE embeddings and ChromaDB for fast vector search. Phase 6 re-ranks candidates with a MiniLM Cross-Encoder. Phase 7 synthesizes answers with Gemini 3.6 Flash at T=0.2 with citations. Phase 8 validates Faithfulness (F >= 0.85) using RAGAS. The UI is built with Streamlit and HTML5 Voice input.'

Q2: Why use RAG instead of Fine-Tuning a Large Language Model?

Answer: 'Fine-tuning modifies model weights, which is expensive and still hallucinates fake verses. RAG keeps the 18 Mahapuranas in a separate database (ChromaDB), allowing the AI to retrieve exact passages and quote real page citations with 100% accuracy.'

Q3: Which embedding model did you choose and why?

Answer: 'I chose BAAI/bge-base-en-v1.5 (768 dimensions). It is a top-ranked open-source model on the Hugging Face MTEB leaderboard, runs 100% locally on CPU without paid API costs, and searches 10,582 chunks in 10 milliseconds.'

Q4: Why is Cross-Encoder re-ranking (Phase 6) necessary after Vector search?

Answer: 'Vector Search is fast for finding 6 candidate passages. The Cross-Encoder performs deep word-by-word cross-attention between the query and passage to pick the Top 3 most accurate passages, cutting out noise and preventing hallucinations.'

Q5: What technologies did you use for the Frontend UI and Voice input?

Answer: 'I used Streamlit with custom CSS to create a minimalist ChatGPT dark-mode interface. For voice input, I integrated the HTML5 Web Speech API directly in the browser so users can click the microphone button and speak their queries for free.'